LG Soft Interview Questions

Technical 1:-

1). Find output

```
char a[100] = “LGSI”;
char *b    = “LGSI”;
printf(“%d  %d ”,sizeof(a), sizeof(b));
```

2) Will it run or not

```
char a[100] = “LGSI”;
char *b    = “LGSI”;
a[2] = ‘P’;
b[2] = ‘Q’;
```

if not than why?

3) Write a code to delete the node at 3rd position in singly linked list & in doubly linked list.
4) Write a code to find loop in the singly linked list.
5)Explain each of them

```
int *p[10];
int  (*p)[10];
int  *(p)[10];
int  **p;
```

6) What is function pointer and write its syntax.
7) find output

```
m[3][3] =  1  2  3
           4  5  6
           7  8  9
```

```
**(m+1)
*(*m + 1)
*(m + 1)
**((m + 1 ) + 1)
*(m + 2) + 1
```

8) Which sorting algorithms do you know?
9) Difference between selection sort and bubble sort. Write algorithm of both, time complexity.
11) Difference Between Semaphore , Mutex and Spinlock?Give example for each, where and how you will use them.
12) Difference between process and threads?
Does main process wait for child process?
13) What does pthread_join()?

14) Write a logic to high the 3rd bit of a number?

Technical 2:-

1) int a = 10;
   void *ptr = &a;
   will it give error or not?

2) int a = 10;
   void *ptr = &a;
   printf(“%d”,*p);
   will it give error or not?
   If yes than what corrections should be made.

3) Write a code to traverse singly linear linked list in backward direction?
4) Write a logic in one line to print the following pattern :

for n = 1	0000000000001
for n = 2	0000000000011
for n = 3	0000000000111
for n = 4	0000000001111
for n = 5	0000000011111
.	.
.	.

& so on

5) What is stucture padding?
Find size of the following and also explain each of them

```
struct xyz
{
        char a;
        int b;
        char c;
        int d;
}

struct xyz
{
        char a;
        int b;
        char c;
}
```

for above que why 12 why not 9?

```
struct xyz
{
        char a;
        double d;
        int b;
```

```
        char c;
}


struct xyz
{
        int a;
        signed int b : 1;
        signed int c : 2;
        signed int d : 29;
}

struct xyz
{
        int a;
        signed int b : 2;
        signed int c : 2;
        signed int d : 29;
}
```

6) Find Output

```
struct xyz
{
        int a;
        signed int b : 2;
        signed int c : 2;
        signed int d : 29;
}stu;
main()
{
        stu.b = 2;
        printf("%d", stu.b);
}
```

7) Convert -1 into binary form.
8) Tell Me about your family background?
9) What kind of field do you like?development or testing?
10) Do you have any problem with location?